

Enfortumab vedotin with pembrolizumab

Enfortumab vedotin (Padcev®) with pembrolizumab (Keytruda®) is a cancer drug treatment. It is used to treat locally advanced or advanced bladder cancer.

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What is enfortumab vedotin (Padcev®) with pembrolizumab Keytruda®)?

Enfortumab vedotin with pembrolizumab is an [antibody drug conjugate](#). This means it is a combination of cancer drugs.

Enfortumab vedotin is also called EV or Padcev®. It is a type of targeted therapy drug called a [monoclonal antibody](#).

Pembrolizumab is also called (Keytruda®). It is a type of [immunotherapy](#) drug called a checkpoint inhibitor.

EV with pembrolizumab may be used as a first treatment to treat:

- [locally advanced bladder cancer](#) that cannot be removed with surgery
- bladder cancer that has spread to other parts of the body, such as the lungs, liver or bones
- [upper urinary tract urothelial cancer \(UTUC\)](#) that cannot be removed with surgery or has spread to other parts of the body.

It is best to read this information with our general information about [muscle-invasive bladder cancer \(MIBC\)](#) and [upper urinary tract urothelial cancer \(UTUC\)](#).

Your cancer team will talk to you about this treatment and its possible side effects before you agree ([consent](#)) to have treatment.

More information about this treatment

This information is correct at time of publishing. But sometimes the types of cancer this treatment is used for, or treatment side effects, may change between revision dates.

You can talk to your cancer team if you want more detailed information about this treatment. Or visit the [electronic Medicines Compendium \(eMC\)](#) website, which has patient information leaflets (PIL) for individual drugs.

Having enfortumab vedotin with pembrolizumab

You have this treatment as a drip into a vein or as an injection under the skin. You usually have it in a day unit or clinic as an outpatient.

During a course of treatment, you will meet someone from your cancer team, such as a:

- cancer doctor
- specialist nurse
- specialist pharmacist.

This is who we mean when we mention doctor, nurse or pharmacist in this information.

During your course of treatment, you will have regular blood tests. This is to check that it is safe for you to have treatment.

You will meet with a doctor, nurse or pharmacist before you have treatment. They will talk to you about your blood results and ask how you have been feeling. If your blood results are okay, the pharmacy team will prepare your treatment.

You have EV first. It is given as a drip into a vein (intravenous infusion) over 30 minutes. You then have pembrolizumab as a drip over 30 minutes or as an injection under the skin. A pembrolizumab injection under the skin is given over a few minutes.

Your nurse will give you the intravenous treatments through one of the following:

- a [cannula](#) – a short, thin tube they put into a vein in the arm or hand
- a [central line](#) – a fine tube that goes under the skin of the chest and into a vein close by
- a [PICC line](#) – a fine tube that is put into a vein in the arm and goes up into a vein in the chest
- an [implantable port](#) (portacath) – a disc that is put under the skin on the chest or arm and goes into a vein in the chest.

Your cancer team may also give you anti-sickness drugs and other medicines to take home. Take these exactly as they tell you to.

Your course of treatment

You usually have EV with pembrolizumab every 3 weeks. Some people have it every 6 weeks.

You may have pembrolizumab for up to 2 years as long as you are not having difficulties with side effects.

If your treatment is working well, you can continue to have EV on its own after pembrolizumab stops.

Your cancer team will discuss your treatment plan with you.

Your cancer team may also give you anti-sickness drugs and other medicines to take home. Take all your capsules or tablets exactly as they tell you to.

About side effects

We explain the most common side effects of this treatment here. We also include some that are less common. This treatment can make the immune system too active. This can cause side effects that may be serious if not treated quickly.

You may get some of the side effects we mention, but you are unlikely to get all of them. And you may have some side effects, including rarer ones, that we have not listed here.

Other cancer treatments may cause different side effects. If you are also having other cancer treatment, you may have other side effects.

Do not try to treat side effects yourself. Always tell your doctor, nurse or pharmacist about any side effects you have. They can give you:

- drugs to help control some side effects
- advice about managing side effects.

It is important to take any drugs exactly as explained. This means they will be more likely to work for you.

Serious and life-threatening side effects

Some cancer treatments can cause serious side effects. Sometimes, these may be life-threatening. Your doctor, nurse or pharmacist can explain the risk of these side effects to you.

Delayed side effects of immunotherapy drugs

Side effects can start during treatment. But some may start weeks, months or sometimes more than a year after you finish treatment.

If you have symptoms or side effects that start during treatment or after treatment ends, contact the hospital straight away on the 24-hour number. Side effects can become serious if they are not treated.

Contact the hospital

Your doctor, nurse or pharmacist will give you 24-hour contact numbers for the hospital. If you feel unwell or need advice, you can call at any time of the day or night. Save these numbers in your phone or keep them somewhere safe.

Your patient alert card

They will also give you a card with information about your treatment. It is important to keep the card with you during treatment and after treatment ends. You should show it to any doctor or healthcare professional caring for you.

Side effects while treatment is being given

Some people have side effects while they are having this treatment or shortly after they have it.

Allergic reaction

Some people have an allergic reaction while having this treatment. Signs of a reaction can include:

- feeling hot or flushed
- shivering
- itching
- a skin rash
- feeling dizzy or sick
- a headache
- feeling breathless or wheezy
- swelling of your face or mouth
- pain in your back, tummy or chest.

Your nurse will check you for signs of a reaction during your treatment. If you feel unwell or have any of these signs, tell them straight away. If you do have a reaction, they can treat it quickly.

Sometimes a reaction happens a few hours after treatment. If you develop any of these signs or feel unwell after you get home, contact the hospital straight away on the 24-hour number.

Always call 999 if swelling happens suddenly or you are struggling to breathe.

The drug leaks outside the vein

Sometimes cancer drugs that are given into a vein may leak outside the vein. If this happens, some drugs can damage the tissue near the vein. This is called extravasation. Extravasation is not common, but it is important that it is dealt with quickly. If you have any of the following symptoms around the vein during or after your treatment, tell your nurse straight away:

- stinging
- pain
- swelling
- if you have white skin, the area may become red
- if you have black or brown skin, the area may become darker.

Very common side effects

These side effects happen to 10 or more people in every 100 people (10% or more) who have this treatment.

We have more information about the side effects of [pembrolizumab](#).

For more information about the side effects of EV, you can visit the electronic [Medicines Compendium \(eMC\) website](#).

Anaemia (low number of red blood cells)

This treatment can reduce the number of red blood cells in your blood. Red blood cells carry oxygen around the body. If the number of red blood cells is low, this is called anaemia. You may feel:

- very low in energy
- breathless
- dizzy and light-headed.

If you have these symptoms, contact the hospital straight away on the 24-hour number. You may need treatment for anaemia. If you are very anaemic, you may need a drip to give you extra red blood cells. This is called a blood transfusion.

Feeling sick

You may feel sick during this treatment, but this is usually mild. Your doctor, nurse or pharmacist may prescribe you anti-sickness drugs to help prevent or control sickness. Take the drugs exactly as they tell you to. [If you feel sick](#), take small sips of fluid often and eat small amounts regularly. It is important to drink enough fluids.

If you continue to feel sick, or if you are sick (vomit) 1 to 2 times in 24 hours, contact the hospital on the 24-hour number as soon as possible. They will give you advice.

Diarrhoea or tummy pain

You may have [diarrhoea](#) or tummy (abdominal) pain with this treatment. Less commonly, it may cause inflammation of the bowel (colitis) which can be very serious.

Diarrhoea means passing more stools (poo) than is normal for you or having watery or loose stools. If you have a stoma, it may be more active than usual.

Contact the hospital straight away on the 24-hour number if you have any of the following symptoms during treatment, or after treatment ends:

- You have diarrhoea 4 or more times in a day.
- You have a moderate increase in stoma activity.
- You have uncomfortable tummy cramps.
- You have diarrhoea at night.
- Your temperature is over 37.5°C.

- There is blood or mucus in your stool.

You may need to go into hospital to have fluids through a drip or treatments such as steroids.

Loss of appetite

This treatment can affect your appetite. Don't worry if you do not eat much for 1 or 2 days. But if your appetite does not come back after a few days, or if you are losing weight, tell your doctor, nurse or pharmacist. They can give you advice. They may give you food or drink supplements. Or they may suggest changes to your diet or eating habits to help.

Changes to your taste

Some foods may taste different or have no taste. Try different foods to find out what tastes best to you. You may also get a bitter or metallic taste in your mouth. Your doctor, nurse or pharmacist can give you advice. It might help to try:

- sucking sugar-free sour or boiled sweets
- eating cold foods
- eating sharp-tasting fresh fruit.

Taste changes usually get better after treatment ends. We have more information about [coping with changes to taste](#).

Weight loss

This treatment may cause weight loss. Let your doctor, nurse or pharmacist know if this happens. They can give you advice.

Feeling tired (fatigue)

[Feeling tired](#) is a common side effect of this treatment. Try to pace yourself and plan your day so you have time to rest. Gentle exercise, like short walks, can help you feel less tired.

Sometimes, this treatment can affect glands that make hormones. This can make you feel tired. Contact the hospital on the 24-hour number if either of these things happen during treatment or after it ends:

- your tiredness gets worse or stops you doing your usual daily activities
- you feel weak or shaky.

If you feel sleepy, do not drive or use machinery.

Hair loss

Your hair may get thinner. But you are unlikely to lose all the hair from your head. Hair loss usually starts after your first or second treatment. It is almost always temporary, and your hair will usually grow back after treatment ends.

Your nurse can talk to you about ways to cope with [hair loss](#).

Numb or tingling hands or feet (peripheral neuropathy)

This treatment may affect the nerves in your fingers and toes. This can cause numbness, tingling or pain in your hands or feet. This is called [peripheral neuropathy](#). You might find it hard to do fiddly tasks such as fastening buttons or tying shoelaces.

If you have these symptoms, always tell your doctor, nurse or pharmacist. They sometimes need to change the drug or the dose of the drug. The symptoms usually improve slowly after treatment ends. But for some people they continue and are a long-term side effect of treatment.

Eyesight changes

This treatment causes dry eyes. Your doctor or nurse can prescribe eye drops to improve this.

Less commonly, other more serious eye changes may happen. Contact the hospital straight away on the 24-hour number if you have any of these symptoms during treatment or after it ends:

- sore, painful or swollen eyes
- eyes that are sensitive to light
- changes to your eyesight
- drooping eyelids.

You may need steroids or other treatments. A specialist eye doctor (ophthalmologist) may also need to check your eyes.

Skin changes

This treatment can cause the following symptoms:

- dry, itchy skin
- raised bumps on the skin
- if you have white skin, a rash that might be red and blotchy
- if you have black or brown skin, a rash that might be darker than surrounding skin
- patches of white or paler skin.

Less commonly, it causes inflammation in the skin. If this is not treated, it can become serious.

You may need creams, steroids or other medicines to treat skin problems. Your doctor may stop your treatment until the skin changes improve.

If you notice any skin changes during treatment, or after treatment ends, contact the hospital on the 24-hour number as soon as possible.

Here are some things you can do to take care of your skin during treatment:

- Wash with lukewarm water and mild, unperfumed, soap-free cleansers.
- Moisturise your skin regularly with unperfumed moisturisers for dry skin conditions.
- Use suncream of at least SPF 30 to [protect your skin from the sun](#). Cover up with clothing and a hat.

Contact the hospital straight away on the 24-hour number if you have any of these symptoms:

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- a rash that is spreading
- blistering or peeling skin
- flu-like symptoms, such as a high temperature and joint pain.

These symptoms can be a sign of a serious skin reaction that needs to be treated immediately in hospital.

Effects on the liver

This treatment may affect how your liver works. It can cause liver inflammation called hepatitis. You will have regular blood tests to check how your liver is working. Tell your doctor, nurse or pharmacist if you have loss of appetite.

Sometimes liver changes can be serious. Contact the hospital straight away on the 24-hour number if you have any of these symptoms during treatment or after it ends:

- yellow skin or eyes
- feeling very sleepy
- dark urine (pee)
- unexplained bleeding or bruising
- pain in the right side of your tummy (abdomen).

If you have black or brown skin, changes in your skin colour may be less noticeable. You might first notice the whites of your eyes look yellow, or that the colour of your urine has changed.

Raised blood sugar levels

This treatment may raise your blood sugar levels and increase the risk of diabetes.

You will have regular blood tests to check this.

Contact the hospital on the 24-hour number if you have these symptoms during treatment or after it ends:

- feeling thirsty all the time
- needing to pass urine (pee) more often than usual
- feeling more tired than usual.
- losing weight without trying.

If you have diabetes, your blood sugar levels may be higher than usual. You may need advice and support from your GP or diabetes team about managing this.

Effects on the lungs

This treatment can cause changes to the lungs. Tell your doctor, nurse or pharmacist if you develop:

- a cough that does not go away
- wheezing
- breathlessness.

You should also tell them if any existing breathing problems get worse. You may have tests to check your lungs.

Hormone changes

This treatment may affect your hormones. Hormones control many different processes in the body.

For example, treatment may affect your thyroid gland so that it does not make enough thyroid hormone. This is called hypothyroidism. It can cause symptoms such as weight gain and tiredness.

You will have regular blood tests to check different hormone levels. Contact the hospital on the 24-hour number if you have any of these side effects during treatment or after it ends:

- increased sweating
- weight gain or weight loss
- dizziness or fainting
- feeling more hungry or thirsty than usual
- loss of sex drive
- passing urine (peeing) more often than usual
- headaches that do not go away
- feeling more tired than usual (fatigue)
- mood changes.

You may need drugs to control your hormone levels.

Other side effects (less common)

These side effects happen to less than 10 in 100 people (less than 10%) who have this treatment. Some of them are much rarer than this but they are still important to know about. Rare means a side effect that happens to less than 1 in 1,000 people (less than 0.1%).

Bruising and bleeding

This treatment can reduce the number of platelets in your blood. Platelets are cells that help the blood to clot.

If the number of platelets is low, you may bruise or bleed easily. You may have:

- nosebleeds
- bleeding gums
- heavy periods
- blood in your urine (pee) or stools (poo)
- tiny red, brown or purple spots that may look like a rash – these spots can be harder to see if you have black or brown skin.

If you have any unexplained bruising or bleeding, contact the hospital straight away on the 24-hour number. You may need a drip to give you extra platelets. This is called a platelet transfusion.

Low number of white blood cells

This treatment can reduce the number of white cells in your blood. These cells fight infection. If you have any signs of an [infection](#), tell your doctor, nurse or pharmacist. These signs include:

- a cough
- a sore throat
- diarrhoea
- needing to pass urine (pee) often, or discomfort when you pass urine.

Contact the hospital straight away on the 24-hour number if you have any of these symptoms:

- a temperature above 37.5°C
- a temperature below 36°C
- you feel unwell, even with a normal temperature.

Muscle and joint pain

This treatment can cause sore or weak muscles. Your joints may also be stiff, sore or swollen. This is called arthritis. If you already have problems with joint pain, this may become worse.

Contact the hospital on the 24-hour number if you have any of these symptoms during treatment or after it ends:

- new or worsening aches or pains
- new or worsening stiffness in a joint
- swollen joints or muscle weakness.

Effects on the heart

This treatment can affect [how the heart works](#). You may have [tests](#) to check how well your heart is working. These may be done before, during and after treatment.

Contact the hospital straight away on the 24-hour number if you have any of these symptoms during or after treatment:

- breathlessness
- dizziness
- changes to your heartbeat (palpitations)
- swollen feet and ankles.
- flu-like symptoms, such as body aches, pains and tiredness.

Other conditions can cause these symptoms, but it is important to get them checked by a doctor.

Always call **999** if you have:

-

chest pain, pressure, heaviness, tightness or squeezing across the chest

- difficulty breathing.

Effects on the pancreas

Rarely pembrolizumab may cause inflammation of the pancreas (pancreatitis). You will have regular blood tests to check your pancreas.

Contact the hospital straight away on the 24-hour telephone number you have been given if you have any of these symptoms:

- severe pain in the centre of your tummy (abdomen)
- feeling or being sick (vomiting)
- indigestion
- a temperature above 37.5°C
- fast heartbeat (tachycardia) or rapid breathing
- yellow skin or eyes – this may be harder to notice if you have black or brown skin
- tenderness or swelling of the tummy.

Effects on the nervous system

Rarely, this treatment can affect the brain and nerves. These effects may be mild at first but can become serious if not treated. Symptoms may include:

- weakness, numbness or tingling in your arms, legs or face
- a headache that does not get better
- seeing double or having drooping eyelids
- difficulty turning or bending your neck
- drowsiness or confusion
- problems with moving or speaking
- fits (seizures).

If you have any of these symptoms during treatment or after it ends, it is important to either:

- contact the hospital straight away on the 24-hour number
- go to the hospital straight away
- call **999**.

Do not drive yourself to hospital.

Other important information

Blood clot risk

Cancer and some cancer treatments can increase the risk of a [blood clot](#). Contact the hospital straight away on the 24-hour number if you have any of these symptoms during or after treatment:

- throbbing pain or swelling in a leg or arm
- reddening of the skin in the area – if you have black or brown skin, this can be harder to notice, but the skin might become darker
- suddenly feeling breathless or coughing.

Always call **999** if you have:

- chest pain
- difficulty breathing.

A blood clot is serious, but it can be treated with drugs called anticoagulants. These thin the blood. Your doctor, nurse or pharmacist can give you more information about preventing and treating blood clots.

Other medicines

Some medicines can affect how this treatment works or be harmful while you are having it. Always tell your cancer doctor, nurse or pharmacist about any drugs you are taking or planning to take, such as:

- medicines you have been prescribed
- medicines you buy in a shop, pharmacy or online
- vitamins or supplements
- [herbal drugs and complementary or homeopathic therapies](#)
- recreational drugs – for example, cannabis.

Vaccinations

Cancer doctors usually recommend that people with cancer have vaccinations for flu and [coronavirus \(covid\)](#). They may also recommend other vaccines, such as Shingrix[®] for shingles. These all help reduce your risk of serious illness from these infections. Most people can have these vaccines, including people with weak immune systems.

You should not have live vaccines if your immune system is weak. This includes if you are having or recently had chemotherapy, radiotherapy or other cancer treatments that affect your immune system. Live vaccines can make you unwell because they contain a very weak version of the illness they protect you against. There are several live vaccines, including the yellow fever vaccine.

It is important to ask your doctor, nurse or pharmacist for advice about having vaccinations. They can explain what vaccines are right for you and when it is best to have them.

Pregnancy

If you have not been through menopause, you will usually be advised to have a pregnancy test a week before you start your treatment.

Contraception

Your doctor, nurse or pharmacist will advise you not to [get pregnant](#) or [make someone pregnant](#) while having this treatment and for some time afterwards. The drugs may harm a developing baby. It is important to use contraception to prevent pregnancy. Follow their advice about:

- what types of contraception to use
- how long after treatment you should continue to use contraception.

Breastfeeding

You are advised not to [breastfeed](#) while having this treatment, or for some time after treatment ends. This is because the drugs could be passed to the baby through breast milk.

Your doctor, nurse or pharmacist can give you more information.

Fertility

Some cancer drugs can affect whether you can [get pregnant](#) or [make someone pregnant](#). If you are worried about this, it is important to talk with your doctor before you start treatment.

Medical and dental treatment

If you need medical treatment for any reason other than cancer, always tell the healthcare professional that you are having cancer treatment. Show them your patient alert card. Give them the contact details for your cancer doctor or cancer team so they can ask for advice.

If you have appointments with a dentist or optician, always tell them you are having cancer treatment. Talk to your cancer team before you have any dental treatment.

About our information

This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer.

References



Visit the [electronic Medicines Compendium \(eMC\)](#) to download a Patient Information Leaflet (PIL) for more detailed information. The leaflet lists all known side effects.

[Cancer Information Development team](#)

Author

Date reviewed

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Our cancer information meets the [PIF TICK](#) quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about [how we produce our information](#).

The language we use

We want everyone affected by cancer to feel our information is written for them.

We want our information to be as clear as possible. To do this, we try to:

- use plain English
- explain medical words
- use short sentences
- use illustrations to explain text
- structure the information clearly
- make sure important points are clear.

We use gender-inclusive language and talk to our readers as 'you' so that everyone feels included. Where clinically necessary we use the terms 'men' and 'women' or 'male' and 'female'. For example, we do so when talking about parts of the body or mentioning statistics or research about who is affected.

You can read more about how we produce our information [here](#).

Get in touch

Macmillan Support Line

The Macmillan Support Line is a free and confidential service for people living with and affected by cancer. If you need to talk, we'll listen.



0300 1000 200

Our Support Line is available Monday to Friday, 9:30am-4:30pm



Email us

Email the Macmillan Fundraising team if you have any questions.

Chat online

Available 8am to 7.45pm, 7 days a week. For confidential, real-time support if you don't want to call.



Online Community

Chat online anonymously to others who understand what you are going through. Our community is available 24/7 and has dedicated forums where you can get advice and ask our experts.



Find cancer support services in your area

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.